

NAME - SAHIL JALAGAR

USN - 2VX25UE404

EXPT NO. - 11

TITLE - MULTI-THREAD APPLICATION

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In [1]: import threading, random, time

number = 0
e1 = threading.Event()
e2 = threading.Event()

def generate():
    global number
    while True:
        number = random.randint(1, 10)
        print("Generated number:", number)
        e1.set()
        time.sleep(1) # every 1 second

def square():
    while True:
        e1.wait()
        print("Square of", number, "=", number**2)
        e1.clear()
        e2.set()

def cube():
    while True:
        e2.wait()
        print("Cube of", number, "=", number**3)
        print("=====")
        e2.clear()

t1 = threading.Thread(target=generate)
t2 = threading.Thread(target=square)
t3 = threading.Thread(target=cube)

t1.start()
t2.start()
t3.start()
```

Generated number: 7

Square of 7 = 49

Cube of 7 = 343

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Generated number: 1

Square of 1 = 1

Cube of 1 = 1

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Generated number: 3

Square of 3 = 9

Cube of 3 = 27

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Generated number: 5

Square of 5 = 25

Cube of 5 = 125

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Generated number: 9

Square of 9 = 81

Cube of 9 = 729

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Generated number: 1

Square of 1 = 1

Cube of 1 = 1

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Generated number: 10

Square of 10 = 100

Cube of 10 = 1000

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